

torch.from_dlpack#

https://pytorch.org/docs/stable/generated/torch.from_dlpack.html#torch.from_dlpack

Description:

Converts a tensor from an external library into a `torch.Tensor`. The returned PyTorch tensor will share the memory with the input tensor (which may have come from another library). Note that in-place operations will therefore also affect the data of the input tensor. This may lead to unexpected issues (e.g., other libraries may have read-only flags or immutable data structures), so the user should only do this if they know for sure that this is fine. `ext_tensor` (object with `__dlpack__` attribute, or a `DLPack` capsule) – The tensor or `DLPack` capsule to convert. If `ext_tensor` is a tensor (or `ndarray`) object, it must support the `__dlpack__` protocol (i.e., have a `ext_tensor.__dlpack__` method). Otherwise `ext_tensor` may be a `DLPack` capsule, which is an opaque `PyCapsule` instance, typically produced by a `to_dlpack` function or method.

Function Signature

`torch.from_dlpack(ext_tensor) → Tensor[source]#`

Parameters

Return type

Returns:

Tensor

Code Examples

```
>>> import torch.utils.dlpack
>>> t = torch.arange(4)

# Convert a tensor directly (supported in PyTorch >= 1.10)
>>> t2 = torch.from_dlpack(t)
>>> t2[:2] = -1 # show that memory is shared
>>> t2
tensor([-1, -1,  2,  3])
>>> t
tensor([-1, -1,  2,  3])

# The old-style DLPack usage, with an intermediate capsule
object
>>> capsule = torch.utils.dlpack.to_dlpack(t)
>>> capsule
<capsule object "dltensor" at ...>
>>> t3 = torch.from_dlpack(capsule)
>>> t3
tensor([-1, -1,  2,  3])
>>> t3[0] = -9 # now we're sharing memory between 3 tensors
>>> t3
tensor([-9, -1,  2,  3])
>>> t2
tensor([-9, -1,  2,  3])
>>> t
tensor([-9, -1,  2,  3])
```

Detailed Documentation

Documentation

Converts a tensor from an external library into a `torch.Tensor`.

The returned PyTorch tensor will share the memory with the input tensor (which may have come from another library). Note that in-place operations will therefore also affect the data of the input tensor. This may lead to unexpected issues (e.g., other libraries may have read-only flags or immutable data structures), so the user should only do this if they know for sure that this is fine.

`ext_tensor` (object with `__dlpack__` attribute, or a `DLPack` capsule) –

The tensor or `DLPack` capsule to convert.

If `ext_tensor` is a tensor (or `ndarray`) object, it must support the `__dlpack__` protocol (i.e., have a `ext_tensor.__dlpack__` method). Otherwise `ext_tensor` may be a `DLPack` capsule, which is an opaque `PyCapsule` instance, typically produced by a `to_dlpack` function or method.

`device` (`torch.device` or `str` or `None`) – An optional PyTorch device specifying where to place the new tensor. If `None` (default), the new tensor will be on the same device as `ext_tensor`.

`copy` (`bool` or `None`) – An optional boolean indicating whether or not to copy self. If `None`, PyTorch will copy only if necessary.

